Describe closed-loop mechatronic system

In which the control action is somehow dependent upon the output and are generally called as feedback control systems.

• Is a system in which the control action depends on the output.

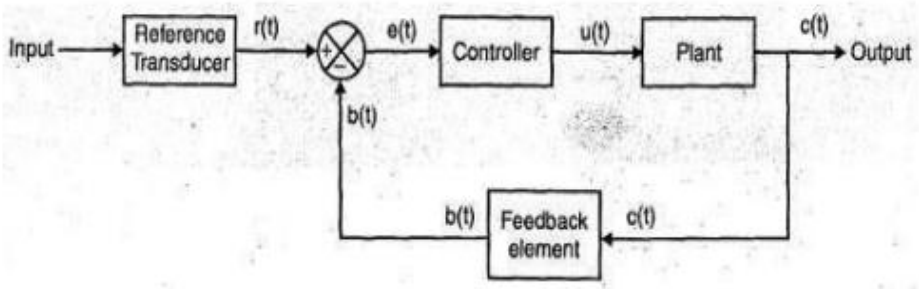
• In closed loop control system, the actuating error signal, which is the difference between the input signal and the feedback signal (output signal or its function) is fed to the controller

Elements of closed loop system are:

1. Command: The command is the externally produced input and independent of the feedback control system.
2. Reference Input Element: It is used to produce the standard signals proportional to the command
3. Error Detector: The error detector receives the measured signal and compares it with reference input. The difference of two signals produces error signal.
4. Control Element: This regulates the output according to the signal obtained from error detector.
5. Controlled System: This represents what we are controlling by feedback loop.
6. Feedback Element: This element feedbacks the output to the error detector for comparison with the reference input.

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Define Mechatronics. List the differences between Open loop and Closed loop systems

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Mechatronics “Synergistic integration of mechanical engineering with electronics and intelligent computer control in the design and manufacturing of industrial products and processes.”

Basis of Difference	Open Loop Control System	Closed Loop Control System
Definition	A control system in which there is no feedback path is provided is called an *open loop control system.*	The control system in which there is a feedback path present is called a *closed loop control system*.
Control action	In open loop control system, the control action is independent of the output of the overall system.	In closed loop control system, the control action is dependent on the output of the system.
Design complexity	The design and construction of an open loop control system is quite simple.	Closed loop control system has comparatively complex design and construction.
Main Components	The major components of an open loop control system are – controller and plant.	The main components of a closed loop control system are – Controller, plant or process, feedback element and error detector (comparator).
Response	Open loop control system has fast response because there is no measurement and feedback of output.	The response of the closed loop control system is slow due to presence of feedback.
Reliability	The reliability of open loop control system is less.	The closed loop control system is more reliable.

Accuracy	The accuracy of open loop control system depends upon the system calibration and therefore, may be less.	Closed loop control system is comparatively accurate because the feedback maintains its accuracy.
Maintenance	Open loop control system requires less maintenance.	Comparatively more maintenance is needed in closed loop control system.
Implementation	Open loop control system is easy to implement.	The implementation of a closed loop control system is relatively difficult.
Cost	Open loop control system is less expensive.	The cost of the closed loop control system is relatively high.
Examples	Common practical examples of open loop control systems are – automatic traffic light system, automatic washing machine, immersion heater, etc.	Examples of closed loop control systems include: ACs, fridge, toaster, rocket launching system, radar tracking system, etc.

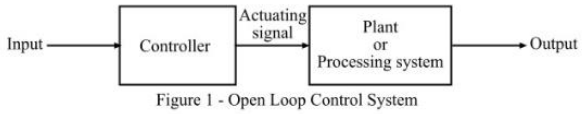


Figure 1 - Open Loop Control System

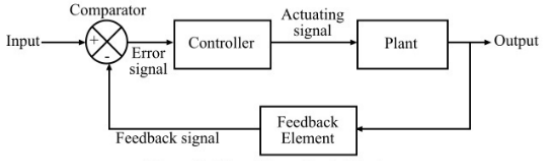
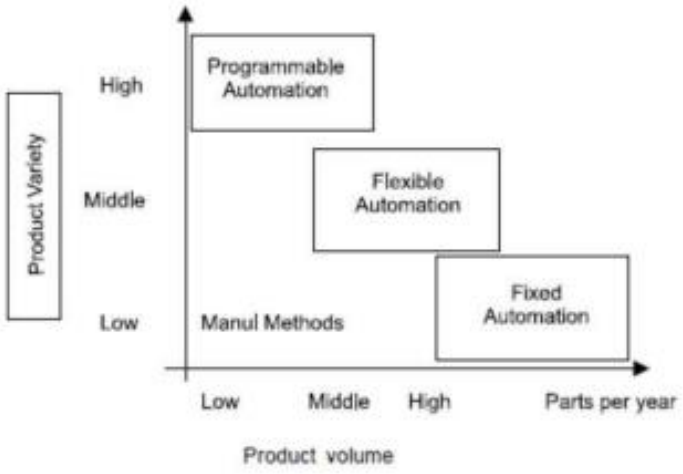


Figure 2 - Closed Loop Control System

Explain the different types of Automation system.
Define Automation. Explain the three types of Automation.

Automation refers to the use of technology to perform tasks or processes without human intervention. It involves the use of machines, software, and systems to control and manage a range of functions, from manufacturing and production to information technology and business operations.



Fixed Automation • Fixed automation, also known as hard automation, is a type of automation system that is designed to perform a specific task or function, typically in a manufacturing or production environment. • Fixed automation systems are highly specialized and cannot be easily adapted to perform other tasks.

Programmable Automation ➢ Programmable automation, also known as soft automation, is a type of automation system that can be programmed or reprogrammed to perform different tasks or functions. ➢ Programmable automation systems are more flexible and adaptable than fixed automation systems, making them well-suited to applications that require frequent changes or adjustments. ➢ These systems typically involve the use of computers and software to control and manage the automation process.

Flexible Manufacturing Flexible automation is a type of automation system that combines the benefits of fixed and programmable automation. ➢ These systems are designed to perform a range of tasks while being customizable and adaptable to changes in production. ➢ Flexible automation systems typically involve the use of robotics, computer controls, and other advanced technologies to enable machines to perform a wide range of tasks, such as sorting, assembly, and packaging

Explain the following robotics configuration: i) Polar cylindrical ii) Cartesian coordinate Based on the configuration, explain the four types of robots with sketches

Cartesian Configuration Robots: ▪ Provides three linear motions along three mutually perpendicular axes: X, Y, and Z. However, there is no rotary motion. ▪ Configuration provides rectangular work envelope. ▪ Used for assembly, palletizing and machine tool loading.

Advantages: 1. Simple controls due to linear movements and easy to visualize 2. High degree of mechanical rigidity, accuracy and repeatability 3. Can carry heavy loads 4. Motion of the wrist end is smoother

Disadvantages: 1. The arm movement is limited to a small rectangular work space 2. It occupies large area there by giving a low ratio robot size to operating volume

Applications: Assembly, welding, machine loading/unloading, surface finishing, inspection, etc.

Cylindrical Configuration Robots:

▪ Provides two linear and one rotary motion. ▪ Configuration provides cylindrical work envelope, has good work area to floor area ratio. ▪ Used for loading and unloading on machine tools.

Advantages: 1. Larger work space than Cartesian configuration 2. Robot is relatively easy to program

Disadvantages: 1. Overall mechanical rigidity is reduced because robots with a rotary axis must overcomethe inertia of the object when rotating 2. Robots repeatability and accuracy is also reduced in the direction of rotary movement 3. It requires more sophisticated control system

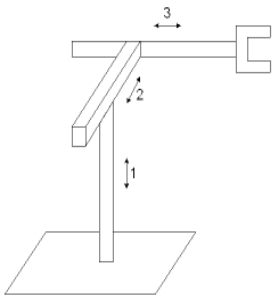
Applications: Pick and place operations, machine loading/unloading, investment casting, forging operations, conveyer pellet transfers, assembly, coating applications, etc.

Polar (Spherical) Configuration Robots: ▪ Provides one linear and two rotary motions. ▪ Configuration provides spherical work envelope. ▪ Used for spot welding and manipulation (Handling) of heavy loads

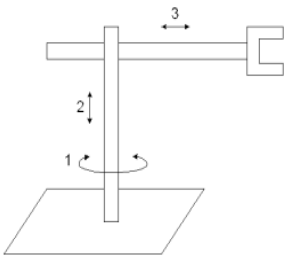
Advantages: 1. It can generate large working envelope compared to Cartesian and cylindrical 2. It is simple in design and easy to program 3. Provides good weight lifting capabilities

Disadvantages: 1. Reduced mechanical rigidity
2. More sophisticated control system is required
3. Limited vertical movement

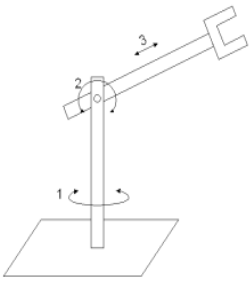
Applications: Die casting, injection moulding, forging, machine tool press loading/unloading, dip coating, material transfer applications, etc.



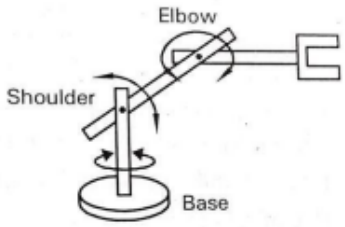
Cartesian: LOO



Cylindrical: TLO



Spherical (Polar): TRL



Jointed Arm Configuration Robot

Jointed Arm Configuration Robot:
It is like that of human arm, consisting of two straight components corresponding to the human forearm and upper arm mounted on a vertical pedestal that can be rotated about the base so as to provide the robot with the capacity to work within a quasi-spherical space. It gives rise to three rotational movements.

Advantages: 1. Most flexible configuration that enables the robot arm to move in all directions 2. Large working envelope 3. Versatile configuration
Disadvantages: 1. Requires complex programming 2. Control actions are different 3. Less stable as the arm approaches maximum reach
Applications: Painting, welding, automatic assembly, welding, pick and place operations involving lightparts, etc

Explain Physical design and Logical design of IoT

Physical Design of IOT

❑ A physical design of an IoT system refers to the individual node devices and their protocols that are utilised to create a functional IoT ecosystem. ❑ Each node device can perform tasks such as remote sensing, actuating, monitoring, etc., by relying on physically connected devices. ❑ It may also be capable of transmitting information through different types of wireless or wired connections. The things/devices in the IoT system are used for: • Building connections • Data processing
• Providing storage • Providing interfaces • Providing graphical interfaces ❑ The devices generate data, and the data is used to perform analysis and do operations for improving the system. ❑ For instance, a moisture sensor is used to obtain the moisture data from a location, and the system analyses it to give an output.

Logical Design of IOT

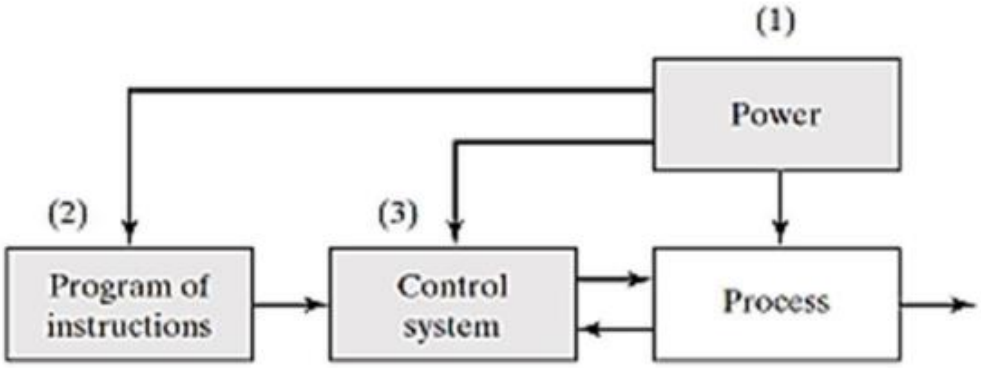
❑ A logical design for an IoT system is the actual design of how its components (computers, sensors, and actuators) should be arranged to complete a particular function. ❑ It doesn't go into the depth of describing how each component will be built with low-level programming specifics

	Explain Functional blocks and communication models used in IoT. **IoT logical design includes:** **IoT functional blocks**: ❑ IoT systems include several functional blocks such as Devices, communication, security, services, and application. ❑ The functional blocks provide sensing, identification, actuation, management, and communication capability. ❑ These functional blocks consist of devices that handle the communication between the server and the host, enable monitoring control functions, manage the data transfer, secure the IoT system using authentication and different functions, and provide an interface for controlling and monitoring various terms. **IoT communications models** There are multiple kinds of models available in an Internet of Things system that is used for communicating between the system and server, such as: • Request-response model • Push-pull model • Publish-subscribe model • Exclusive pair model IoT communication APIs APIs are used to communicate between the server and system in IoT. Some API's include: • REST-based communication APIs • Client-server • Stateless • Cacheable • Websocket based communication API IoT Protocols ❑ The set of rules governing all direct or indirect exchange of data between computers on a network. ❑ These rules are formulated at the application level and are used collectively to define how devices communicate interoperably, irrespective of differences in their internal designs and operations. ❑ **IoT protocols** help send commands and data between a network of devices controlled by sensors or other physical attributes like motion, temperature, or vibration. ❑ Network protocols help achieve reliable data transfer across all layers like the application, transport, network, and link layer.
	Describe the basic elements of automation system with block diagram Describe the basic elements of automation system with block diagram The basic elements of automation are: ➢ **Sensing Devices**: These are devices that can detect changes in the environment, such as temperature, pressure, or motion, and convert them into electrical signals that can be processed by a computer or control system.

➢ **Control Devices**: These are devices that receive input from the sensing devices and use that information to control the operation of machines or equipment, such as motors, valves, or actuators.

➢ **Actuators:** These are devices that convert electrical signals from the control system into physical motion or force, such as hydraulic or pneumatic cylinders, or electric motors. Power Source: This is the energy source that provides power to the machines or equipment being controlled, such as electricity or compressed air

➢ **Control System**: This is the software or hardware that controls the operation of the machines or equipment, and processes data from the sensing devices and control devices to make decisions about how to control the system.



Define IoT. List the characteristics of IoT.

The Internet of Things (IoT) describes the network of physical objects—"things"—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.

Characteristics of IoT

Characteristics are the reason behind the success of IoT. There are six main characteristics. Each characteristic encompasses a set of capabilities that make IoT a success.

Intelligence IoT systems are extensively liked in the market because of their intelligence. A combination of algorithms and computer enables the system to inform change in the environment and take appropriate actions. For example – Systems are intelligent enough to sense a sudden spike in temperature and trigger an alarm for fire.

Connectivity

Connectivity is the main characteristic of IoT as it enables the system to send data and stay connected to other devices. It provides system network accessibility and function collaboratively.

Expressing IoT is all about interacting intelligently with the outer environment and humans. Expressing enables this interactivity. Expressing allows us to show output into the real world and input from people and the environment.

Sensing Sensitivity means aware of the changes around us. Sensor technologies provide us with the means to create an experience that reflects an awareness of changes in the physical world and the people in it. It helps in expressing. This forms the input for the IoT system and provides a better understanding of the complex world around us.

Energy Everything in this world is driven by energy. IoT systems are created smart enough to synthesize energy from the outer environment and conserve it. It is also made energy efficient to work for a longer duration.

Security Safety and security is the most important feature of any system. If the system is not secure to cyber attack and illegal intervention, nobody will use it. IoT systems deal with personal data; that's why it's an obligation that all safety measure should be taken care of in this system. All IoT systems are secure enough to deal with personal data

